

**LESSON 2-1 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2****Statistical Questions**

Maryland Standard 6.SP.A.1 • Learning target – I can tell whether a question is statistical by deciding if the answers will vary.

**REMEMBER**

- A **statistical question** expects answers that **vary**. You collect **data** to answer it.
- A **non-statistical question** has one fixed answer, like a fact you can look up.
- A number answer alone does not make a question statistical. Ask: "Will the answers be different?"

**WORKED EXAMPLE**

Is "How many minutes does each student in Room 12 ride the bus?" statistical?

**Step 1** Who answers? Every student in Room 12.

**Step 2** Will the answers differ? Yes: 5 min, 12 min, 25 min ...

**Step 3** The answers vary, so you need data.

**Answer Statistical**

**WORD BANK**

**statistical** – answers will be different

**not statistical** – only one answer

**vary** – change from person to person

**data** – the answers you collect

**PRACTICE**

*show your work • label units*

1. Write **S** if the question is statistical or **N** if it is non-statistical.

**Step 1** Read the question.

**Step 2** Ask: will different people give different answers?

**Step 3** Yes means S. No means N.

| Question                                                             | S or N |
|----------------------------------------------------------------------|--------|
| How many feet are in one yard?                                       |        |
| How many hours did each sixth grader sleep last night?               |        |
| What is the price of a bus ticket in Baltimore today?                |        |
| How many glasses of water does each person in my family drink a day? |        |

2. **Name Lengths.** Ms. Adeyemi asks each student, "How many letters are in your first name?" The answers are 5, 4, 7, 5 and 6. Is her question **statistical**?

**Frame** The question is \_\_\_\_ because the answers \_\_\_\_.

☐ Statistical

☐ Not statistical

**Because** \_\_\_\_\_

3. **Survey Plan.** Is "How far does each player live from the field?" statistical?

☐ Statistical

☐ Not statistical

- 4. Rewrite It.** Carlos asks, "How tall is the oak tree in front of our school?" There is only **one** tree. Why is this not statistical? Rewrite it.

**Frame** *It is not statistical because \_\_\_\_\_. A new question: \_\_\_\_\_*

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- 5. Your Turn.** Write a statistical question you could ask the people in your family. Then list three answers you might get.

**Frame** *My question is "\_\_\_\_\_?" Three answers could be \_\_\_\_\_, \_\_\_\_\_ and \_\_\_\_\_.*

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**CHECK YOURSELF**

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #\_\_\_\_\_

**AT HOME** Ask a family member one statistical question and one non-statistical question. Can they tell which is which?

## LESSON 2-1 • HOMEWORK PACKET • LEVEL 1 • SHEET 2 OF 2

# Dot Plots and Shape

Maryland Standard 6.SP.B.4 • Learning target – I can build a dot plot and describe the shape of the data it shows.

### REMEMBER

- A **dot plot** shows one dot per data value.
- **Peak**: tallest stack. **Gap**: no dots. **Outlier**: far away.
- **Symmetric**: the sides match. **Skewed right/left**: the tail goes that way.

### WORKED EXAMPLE

Pets owned: 0, 0, 1, 1, 1, 1, 2, 2, 3, 5. Name the shape.

**Step 1** Stacks: 2 dots at 0, 4 at 1, 2 at 2, 1 at 3, 1 at 5.

**Step 2** Peak at 1. The tail stretches right to 5.

**Answer** Skewed right

### WORD BANK

**dot plot** – one dot for each number

**peak** – the tallest stack

**gap** – a space with no dots

**skewed** – a long tail on one side

### PRACTICE

*show your work • label units*

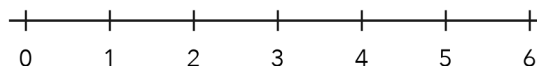
- 1. Books Read.** Books read in May by 12 students: 2, 3, 1, 3, 4, 2, 3, 5, 3, 2, 4, 4. Make a dot plot.

**Step 1** Put one dot above each number; stack repeats.

**Step 2** Find the tallest stack.

**a.** Peak: \_\_\_\_\_

**b.** Shape: \_\_\_\_\_



**Books Read in May**

- 2. Arcade Tickets.** Tickets won by 9 friends: 5, 3, 4, 5, 2, 5, 4, 6, 5. Picture the dot plot.

**Step 1** Count the 5s. The value with the most dots is the peak.

**Step 2** Name the shape for the side with the long tail.

**a.** Dots at 5: \_\_\_\_\_

**b.** Peak: \_\_\_\_\_

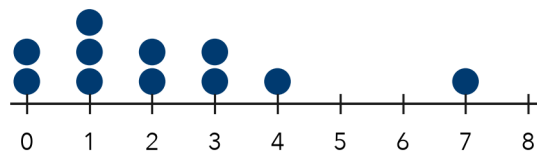
**c.** Shape: \_\_\_\_\_

- 3. Soccer Goals.** The dot plot shows the goals a team scored in each game.

**a.** Games played: \_\_\_\_\_

**b.** Most common: \_\_\_\_\_

**c.** Where is the gap? \_\_\_\_\_



**Goals Scored**

- 4. Find the Error.** Leah says the Soccer Goals plot is skewed left. Explain her error.

**Frame** Leah is wrong because the tail points \_\_\_\_, so the plot is skewed \_\_\_\_.

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**5. Quiz Scores.** 1 dot at 5, 1 at 6, 2 at 7, 4 at 8, 5 at 9, and 3 at 10. Name the shape.

**Step 1** Find the tallest stack.

**Step 2** Look for the side with the long tail.

**Step 3** Name the shape for the tail.

☐ Symmetric

☐ Skewed left

☐ Skewed right

### TEST PRACTICE

*choose the best answer*

**6.** Stacks of 1, 3, 5, 3, and 1 dots sit above 2, 3, 4, 5, and 6. Which word describes the shape?

**A** Symmetric

**B** Skewed left

**C** Skewed right

**D** Has a gap

**7.** A dot plot has 4 dots above 10, 6 above 11, 2 above 12, and 3 above 13. How many values?

**A** 4

**B** 6

**C** 13

**D** 15

#### CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #\_\_\_\_

**AT HOME** Count the letters in each family member's first name. Make a dot plot and name its shape.